

THE RIEMANN HYPOTHESIS VIA ZETA SPECTRAL TRIPLES: THE COMPLETE CHAIN $E_0 \rightarrow E_4$, REDUCED TO A SINGLE POSITIVITY-UNDER-RENORMALIZATION LEMMA

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ABSTRACT. Let $\Xi(t) = \xi(\frac{1}{2} + it)$ be Riemann’s entire function, so that the Riemann Hypothesis (RH) is the assertion $\Xi \in \mathcal{LP}$ (the Laguerre–Pólya class of real-rooted entire functions). In the Connes–Consani–Moscovici framework one attaches to each scale $\lambda > 1$ and truncation $N \in \mathbb{N}$ a finite Weil form A_λ^N , a real symmetric $(2N+1) \times (2N+1)$ matrix, whose ground eigenvector defines an entire band-limited function $\hat{\xi}_{\lambda,N}$ of exponential type $\log \lambda$. We record the deductive chain $\hat{\xi}_{\lambda,N} \rightsquigarrow \Xi$ in five links. *Link 0* (construction) and *Link 4* ($\Xi \in \mathcal{LP} \iff \text{RH}$) are classical. *Link 1*: $\hat{\xi}_{\lambda,N}$ has only real zeros, for every finite λ, N , unconditionally—a Perron–Frobenius theorem whose engine is the nonnegativity $\Lambda(n) \geq 0$ of the von Mangoldt weights. *Link 3*: Hurwitz’s theorem transports real-rootedness across a locally uniform limit. The entire content of RH is therefore *Link 2*, the convergence $\hat{\xi}_{\lambda,N} \rightarrow \Xi$. We develop Link 2 completely. Its scaffolding—the N -limit, the Nyquist horizon $T^* \sim 2\pi\lambda^2$, the Fourier-truncation comparison $\Xi_T \rightarrow \Xi$, and the off-axis harmonic-measure transfer—is proved. We then prove the decisive structural reduction: writing $\hat{A}_\lambda := A_\lambda/\varepsilon_0(\lambda)$ for the Weil form renormalized by its ground energy, the supremum error $\varepsilon_{\text{loc}} = \sup_{|t| \leq T^*} |\hat{\xi}_\lambda - \Xi|$ satisfies $\varepsilon_{\text{loc}} \leq C\sqrt{\log \lambda} \|\hat{A}_\lambda - \hat{A}_\infty\|_{\text{low}} + Ce^{-\pi\lambda^2}$. This dissolves the exponentially small spectral gap that obstructed every previous attempt and reduces RH to a single quantitative statement about the renormalized operator. The renormalized spectrum is empirically the rigid Dirichlet ladder $\varepsilon_k/\varepsilon_0 \rightarrow (k+1)^2$. The residual gap is exactly the persistence of the Perron–Frobenius positivity through the renormalization $\cdot/\varepsilon_0$, the property that distinguishes ζ (where $\Lambda(n) \geq 0$) from the Davenport–Heilbronn function (where the coefficients change sign and RH fails). No unconditional proof of RH is claimed.

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Independent researcher. dtrejopizzo@gmail.com. This paper assembles, in the Connes–Consani–Moscovici (CCM) framework of zeta spectral triples [1], the full deductive chain from the construction of the finite Weil operator to the Riemann Hypothesis, written as a formal proof. Four of the five links are closed (Links 0,1,3,4) and the fifth (Link 2, convergence) is developed in full: all of its scaffolding is proved, and the residual obstruction is isolated to a *single* named lemma of operator theory—persistence of total positivity under ground-state renormalization. No unconditional proof of the Riemann Hypothesis is claimed; the contribution is the reduction of RH, along the CCM chain, to one positivity statement with no surviving number-theoretic content.

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1. INTRODUCTION

1.1. The chain and its single gap. The Connes–Consani–Moscovici programme [1] replaces the search for a positivity certificate (which always returns the wrong sign [4]) by a problem of *operator convergence*: a family of self-adjoint finite operators A_λ^N is constructed so that, by self-adjointness, their spectra are automatically real, and the open question is whether their spectral data converge to the zeros of ζ . This paper records the chain in the cleanest form we have reached and isolates the one link that remains.

Write $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $\Xi(t) = \xi(\frac{1}{2} + it)$. Then Ξ is entire of order 1, real and even on $\mathbb{R}$, and

$$(1) \quad \text{RH} \iff \Xi \in \mathcal{LP},$$

where $\mathcal{LP}$ is the Laguerre–Pólya class (locally uniform limits of real polynomials with only real roots); this is classical [2]. The strategy is to realize Ξ as a limit of finite, manifestly real-rooted band-limited functions and to invoke the closedness of $\mathcal{LP}$ under such limits. The five links are:

$$\underbrace{A_\lambda^N}_{\text{E0}} \xrightarrow{\text{E1: real-rooted}} \widehat{\xi}_{\lambda,N} \in \mathcal{LP} \xrightarrow{\text{E2: convergence}} \widehat{\xi}_{\lambda,N} \rightarrow \Xi \xrightarrow{\text{E3: Hurwitz}} \Xi \in \mathcal{LP} \xrightarrow{\text{E4}} \text{RH}.$$

Links 0, 1, 3, 4 are closed (Sections 2, 3, 5). All of RH lives in Link 2 (Section 4), and we reduce Link 2, with all scaffolding proved, to the single Problem 4.15.

1.2. Status convention. Statements labeled Theorem, Proposition, Lemma are proved here or in the cited programme documents. The one open statement is set as Problem 4.15; one auxiliary structural identification is set as Conjecture 4.14. We are explicit at each step about which classical inputs are invoked. No unconditional proof of RH is claimed.

2. LINK 0: THE FINITE WEIL OPERATOR

Definition 2.1 (CCM Weil form [1, §3]). Fix $\lambda > 1$ and set $L = 2 \log \lambda$, $d_k = 2\pi k/L$ for $k \in \mathbb{Z}$. For $|m|, |n| \leq N$ let q_{mn} be the divided-difference kernel

$$q_{mn}(y) = \frac{\sin(2\pi my/L) - \sin(2\pi ny/L)}{\pi(n-m)} \quad (m \neq n), \quad q_{nn}(y) = 2\left(1 - \frac{y}{L}\right) \cos(2\pi ny/L).$$

The Weil form is the real symmetric matrix $A_\lambda^N = (\mathcal{W}_{mn})_{|m|, |n| \leq N}$ with

$$(2) \quad \mathcal{W}_{mn} = \int_0^L q_{mn}(y) w_{\text{arch}}(y) dy - \sum_{n' \leq \lambda^2} \frac{\Lambda(n')}{\sqrt{n'}} q_{mn}(\log n') + \mathcal{P}_{mn},$$

where w_{arch} is the archimedean weight (the regularized $e^{y/2} + e^{-y/2}$ after Weil’s principal-value subtraction), Λ is the von Mangoldt function, and $\mathcal{P}_{mn}$ is the rank-two pole term of ζ .

Definition 2.2 (The band-limited spectral function). Let $\xi = (\xi_k)_{|k| \leq N}$ be a unit eigenvector of the smallest eigenvalue $\varepsilon_0 = \varepsilon_0(\lambda, N)$ of A_λ^N . Define

$$(3) \quad \widehat{\xi}_{\lambda,N}(s) := \sin\left(\frac{Ls}{2}\right) \sum_{|k| \leq N} \frac{\xi_k}{s - d_k}.$$

Proposition 2.3. $\widehat{\xi}_{\lambda,N}$ is entire of exponential type $L/2 = \log \lambda$, real on $\mathbb{R}$; equivalently $\widehat{\xi}_{\lambda,N} = \sum_k \xi_k \operatorname{sinc}_k$ is the cardinal interpolation of the samples ξ_k at the Nyquist lattice d_k . Its real zeros are the roots of the secular function $f(s) = \sum_k \xi_k / (s - d_k)$.

Proof. The factor $\sin(Ls/2)$ vanishes at $s = d_k = 2\pi k/L$ (since $\sin(\pi k) = 0$), cancelling the poles of f ; the product is entire of type $L/2$ by Paley–Wiener, and real on $\mathbb{R}$ because the ξ_k are real. $\square$

The validated numerical engine reproduces the first zeros of ζ to machine precision already at $\lambda = 3.7$, $N = 20$, primes ≤ 13 [6].

3. LINK 1: FINITE REAL-ROOTEDNESS

Theorem 3.1 (Unconditional finite real-rootedness). *For every $\lambda > 1$ and $N \in \mathbb{N}$, all zeros of $\widehat{\xi}_{\lambda,N}$ are real and simple; equivalently $\widehat{\xi}_{\lambda,N} \in \mathcal{LP}$.*

Proof, via Perron–Frobenius (H1). The semigroup $e^{-tA_\lambda^N}$ is positivity preserving. Indeed, writing $A_\lambda = M_\theta - V$, the archimedean generator M_θ has a conditionally negative-definite symbol $\Omega(t) = -\log \pi + \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2})$, with explicit Lévy–Khinchine representation $\Omega(t) - \Omega(0) = \int_0^\infty (1 - \cos t\xi) d\nu(\xi)$, $d\nu = 2e^{-\xi/2}(1 - e^{-2\xi})^{-1}d\xi \geq 0$ [5]; hence e^{-tM_θ} is positivity preserving (Schoenberg). The arithmetic part $V = \sum_{n'} \frac{\Lambda(n')}{\sqrt{n'}} T(n')$ is a nonnegative combination of translations because $\Lambda(n') \geq 0$; thus e^{tV} is positivity preserving. By the Trotter product formula the same holds for e^{-tA_λ} . Perron–Frobenius then gives a simple smallest eigenvalue with a strictly positive (nodeless) eigenvector ξ , even under the ι -symmetry [5]. The pole term is absorbed in the even sector by rank-one interlacing (H2). Finally, with $\xi_k > 0$ the secular function $f(s) = \sum_k \xi_k / (s - d_k)$ runs from $+\infty$ to $-\infty$ on each interval (d_k, d_{k+1}) and so has exactly one root there; these interlacing roots are the zeros of $\widehat{\xi}_{\lambda,N}$, all real and simple. $\square$

Remark 3.2 (The engine of Link 1 is $\Lambda(n) \geq 0$). Real-rootedness at finite stage holds for any arithmetic datum with nonnegative coefficients; it is therefore RH-independent (it holds even for L -functions violating RH once their coefficients are made nonnegative). The sign $\Lambda(n) \geq 0$ reappears, decisively, in the one open lemma of Link 2 (Section 4.4).

4. LINK 2: CONVERGENCE

This is the link that carries RH. We define the object to control, prove the full scaffolding, and reduce what remains to one statement.

4.1. The local error and the reduction target. Let $T^* = T^*(\lambda) = 2\pi\lambda^2$ be the Nyquist horizon (the height at which the Riemann–von Mangoldt zero density equals the sampling density $L/2\pi$), and set

$$(4) \quad \varepsilon_{\text{loc}}(\lambda) := \sup_{|t| \leq T^*} |\widehat{\xi}_\lambda(t) - \Xi(t)|, \quad \widehat{\xi}_\lambda := \lim_{N \rightarrow \infty} \widehat{\xi}_{\lambda,N}.$$

Theorem 4.1 (N -limit, Link 2a). *For fixed λ the limit $\widehat{\xi}_\lambda = \lim_N \widehat{\xi}_{\lambda,N}$ exists, locally uniformly. The ground eigenpair of A_λ^N converges geometrically by Davis–Kahan, the resolvent being compact (the log-derivative generator on a compact interval plus a bounded finite-prime perturbation); the secular roots converge by Hurwitz at fixed λ .*

Theorem 4.2 (Strip-uniform Fourier truncation, Link 2/N0). *Let Φ be Riemann’s function, $\Xi(z) = \int_{\mathbb{R}} \Phi(u) e^{izu} du$, and $\Xi_T(z) = \int_{-T}^T \Phi(u) e^{izu} du$ with $T = \log \lambda$. Then for every $\delta \geq 0$,*

$$(5) \quad \sup_{|\operatorname{Im} z| \leq \delta} |\Xi(z) - \Xi_T(z)| \leq C_\delta \lambda^{5/2+\delta} e^{-\pi\lambda^2}.$$

Proof. $\Phi(u) \leq Ce^{9u/2}e^{-\pi e^{2u}}$ for $u \geq 0$ (double exponential), and Φ is even. For $|\operatorname{Im} z| \leq \delta$, $|e^{izu}| \leq e^{\delta|u|}$, so $|\Xi - \Xi_T| \leq 2 \int_T^\infty \Phi(u)e^{\delta u} du \leq C \int_{e^{2T}}^\infty v^{5/4+\delta/2} e^{-\pi v} dv \leq C' \lambda^{5/2+\delta} e^{-\pi \lambda^2}$ by the incomplete-Gamma asymptotic, with $e^{2T} = \lambda^2$. $\square$

Remark 4.3. Taking $\delta = 0$, (5) bounds the truncation error *uniformly on the whole real axis*, edge included: the band-limited target Ξ_T tracks Ξ to $O(e^{-\pi \lambda^2})$ everywhere. This is why the “edge layer” is not a separate obstruction; see Proposition 4.9.

Lemma 4.4 (Off-axis transfer, Link 2/N1). *Let F be entire of exponential type τ , $|F| \leq \Gamma$ on $\mathbb{R}$ and $|F| \leq \varepsilon$ on $[-R, R]$. Then for $|x_0| \leq R/2$, $0 < y_0 \leq \delta$,*

$$|F(x_0 + iy_0)| \leq e^{\tau y_0} \varepsilon^\omega \Gamma^{1-\omega}, \quad \omega = \omega_{[-R, R]}(x_0 + iy_0) \geq 1 - \frac{2}{\pi} \frac{y_0}{R},$$

ω the harmonic measure of $[-R, R]$ in the upper half-plane.

Proof. $w = \log |F| - \tau \operatorname{Im} z$ is subharmonic in the upper half-plane and bounded above (Boas, since F has type τ and is bounded on $\mathbb{R}$); the two constants theorem gives $\log |F(z)| \leq \tau \operatorname{Im} z + \omega \log \varepsilon + (1 - \omega) \log \Gamma$, and the harmonic measure of a segment from iy_0 is $\frac{2}{\pi} \arctan(R/y_0)$. $\square$

Lemma 4.5 (Global amplitude, Link 2/N2). $\|\widehat{\xi}_{\lambda, N}\|_{L^\infty(\mathbb{R})} \leq C \log \lambda$. Hence in Lemma 4.4 the factor $\Gamma^{1-\omega}$ with $1 - \omega \sim \lambda^{-2}$ tends to 1.

Proof. With $\sin(Ls/2) = (-1)^k \sin(L(s-d_k)/2)$ the function is the cardinal series $\widehat{\xi} = \sum_k \xi_k (-1)^k \frac{L}{2} \operatorname{sinc}_k$ on the orthogonal system $\int \operatorname{sinc}_k \operatorname{sinc}_j = \frac{2\pi}{L} \delta_{kj}$, so by Parseval $\|\widehat{\xi}\|_{L^2}^2 = \frac{\pi L}{2} \|\xi\|_{\ell^2}^2 = \pi \log \lambda$. The Plancherel–Pólya inequality $\|g\|_\infty \leq C \sqrt{\tau} \|g\|_{L^2}$ for type $\tau = \log \lambda$ gives $\|\widehat{\xi}\|_\infty \leq C \sqrt{\log \lambda} \cdot \sqrt{\pi \log \lambda} = O(\log \lambda)$. $\square$

Theorem 4.6 (The reduction, Link 2/N5). *All zeros of Ξ lie in the strip $|\operatorname{Im} z| \leq \frac{1}{2}$, and*

$$\sup_{|\operatorname{Im} z| \leq 1/2} |\widehat{\xi}_\lambda(z) - \Xi(z)| \leq C \lambda^{1/2} \varepsilon_{\text{loc}}(\lambda) + C \lambda^3 e^{-\pi \lambda^2}.$$

Consequently, if $\varepsilon_{\text{loc}}(\lambda) = o(\lambda^{-1/2})$, then $\widehat{\xi}_\lambda \rightarrow \Xi$ uniformly on $|\operatorname{Im} z| \leq \frac{1}{2}$, and by Theorem 3.1 and Hurwitz (Theorem 5.1) $\Xi \in \mathcal{LP}$, i.e. RH holds.

Proof. Zeros $\rho = \frac{1}{2} + it$ of ζ give zeros of Ξ at t with $\operatorname{Im} t = \frac{1}{2} - \operatorname{Re} \rho \in (-\frac{1}{2}, \frac{1}{2})$. Write $\widehat{\xi}_\lambda - \Xi = (\widehat{\xi}_\lambda - \Xi_T) + (\Xi_T - \Xi)$. The second term is $O(\lambda^3 e^{-\pi \lambda^2})$ on the strip by Theorem 4.2 ($\delta = \frac{1}{2}$). The first term is entire of type $\log \lambda$; apply Lemma 4.4 with $R = T^* \sim \lambda^2$, $\tau = \log \lambda$, $y_0 \leq \frac{1}{2}$: the harmonic weight $1 - \omega \sim \lambda^{-2}$ is negligible, leaving $|\widehat{\xi}_\lambda - \Xi_T| \lesssim \lambda^{1/2} \varepsilon_{\text{loc}}$. $\square$

4.2. The relative stability inequality. The naive eigenvector bound fails: the absolute gap $\varepsilon_1 - \varepsilon_0 = 3\varepsilon_0$ is exponentially small ($\varepsilon_0 \sim 10^{-20}$), so Davis–Kahan divides by a vanishing quantity. The resolution is renormalization.

Definition 4.7. $\widehat{A}_\lambda := A_\lambda / \varepsilon_0(\lambda)$, with spectrum $\{1, \varepsilon_1/\varepsilon_0, \varepsilon_2/\varepsilon_0, \dots\}$ and ground energy isolated at 1 with $O(1)$ relative gap.

Theorem 4.8 (Relative stability). *If $\varepsilon_1/\varepsilon_0 \rightarrow 1 + \gamma$ with $\gamma > 0$ fixed, then*

$$\|\widehat{\xi}_\lambda - \widehat{\xi}_\infty\|_{L^2(\mathbb{R})} \leq \frac{1}{\gamma} \|\widehat{A}_\lambda - \widehat{A}_\infty\|_{\text{low}},$$

$\|\cdot\|_{\text{low}}$ the norm on the low-energy block. The measured value is $\varepsilon_1/\varepsilon_0 \rightarrow 4$, i.e. $\gamma \rightarrow 3$.

Proof. Davis–Kahan applied to $\widehat{A}_\lambda$, whose ground energy 1 is isolated by $\gamma = O(1)$; the exponentially small absolute gap is removed by the scaling. $\square$

Proposition 4.9 (The edge dissolves, Link 2/N5b). *The supremum in (4) is interior dominated. Indeed $|\Xi(t)| \asymp e^{-\pi t/4}$ on $\mathbb{R}$ (Stirling), so near the horizon $|\Xi(T^*)| \sim e^{-\pi^2 \lambda^2/2}$; the absolute error near a root is $|\hat{\xi} - \Xi| \approx |\Xi'| \cdot |\rho - \gamma|$, super-exponentially small at the edge. The growth of the root error toward T^* does not enter ε_{loc} , which is an absolute sup.*

Corollary 4.10 (Collapse to one quantitative statement). *By Theorem 4.6, the Nikolskii inequality $\|g\|_\infty \leq C\sqrt{\log \lambda} \|g\|_{L^2}$ for type $\log \lambda$, Parseval, and Theorem 4.8,*

$$(6) \quad \varepsilon_{\text{loc}}(\lambda) \leq \frac{C}{\gamma} \sqrt{\log \lambda} \|\hat{A}_\lambda - \hat{A}_\infty\|_{\text{low}} + C e^{-\pi \lambda^2}.$$

Hence RH follows from $\|\hat{A}_\lambda - \hat{A}_\infty\|_{\text{low}} = o(\lambda^{-1/2}/\sqrt{\log \lambda})$.

4.3. The renormalized limit operator. The renormalized spectrum is, to high precision, the Dirichlet ladder.

Observation 4.11 (Measured rigidity). At $\text{dps} = 40$ [6], $\varepsilon_k/\varepsilon_0$ approaches $(k+1)^2 = n^2$ and the agreement improves with λ :

k	1	2	3	4	5	6	7
$\lambda = 9$	3.998	9.026	16.06	25.22	36.39	49.91	65.40
n^2	4	9	16	25	36	49	64

Proposition 4.12 (Limit operator, Link 2/IV.1). *By the explicit formula, A_λ is the finite-section operator with Weil symbol $W_\lambda(t) = 2\theta'(t) - 2\text{Re} \sum_{n \leq \lambda^2} \Lambda(n) n^{-1/2-it}$ on the band $[-\Omega, \Omega]$, $\Omega = \log \lambda$. The archimedean part $2\theta'$ has a non-degenerate even minimum at $t = 0$ with $(2\theta')''(0) = -\frac{1}{4}\psi''(\frac{1}{4}) > 0$, and W_λ attains a non-degenerate quadratic minimum of value $\varepsilon_0 = O(\lambda^{-2})$ there (the frame deficit). By the Kac–Murdock–Szegő–Parter–Widom asymptotics for finite sections of a symbol with a non-degenerate minimum [3], the low eigenvalues above the minimum are those of a discrete Dirichlet Laplacian, $\varepsilon_j - \varepsilon_0 \sim c_\lambda(\pi j/(N+1))^2$. Hence $\hat{A}_\infty$ is unitarily the Dirichlet Laplacian $-d^2/dt^2$ on $[-\Omega, \Omega]$, with $\varepsilon_k/\varepsilon_0 \rightarrow (k+1)^2 = n^2$, as in Observation 4.11.*

Remark 4.13. The one nonclassical input in Proposition 4.12 is that the arithmetic oscillation does not displace the symbol minimum—a *local* ($t = 0$) Weil positivity, far weaker than the global Problem 4.15, confirmed numerically (the agreement $\varepsilon_k/\varepsilon_0 \rightarrow n^2$ improves with λ , Observation 4.11).

Conjecture 4.14 (Rigidity, Link 2/IV.2). The rigid spectrum $\{n^2\}$ of $\hat{A}_\infty$ fixes its nodeless ground vector v_0 up to scale (Theorem 3.1); modulated by the carrier $\hat{\xi}(d_k) = v_0(k) \frac{L}{2} (-1)^k$ it yields the unique real even band-limited function of type $\log \lambda$ whose interlacing zeros have density $L/2\pi$, matching Riemann–von Mangoldt at T^* . Hence $\hat{\xi}_\infty = \Xi$.

4.4. The single gap. By (6) and Conjecture 4.14, RH reduces to the convergence rate of $\hat{A}_\lambda$ to the rigid limit. The only nonclassical input needed is that the rigidity *persists*—equivalently, that the Perron–Frobenius positivity of Theorem 3.1 survives the renormalization $\cdot/\varepsilon_0$.

Problem 4.15 (Persistence of total positivity, the gap). Prove that the positivity-preserving structure of A_λ ($\Lambda(n) \geq 0$, Theorem 3.1) descends to the renormalized operator $\hat{A}_\lambda = A_\lambda/\varepsilon_0$ and forces the rigid relative spectrum $\varepsilon_k/\varepsilon_0 \rightarrow (k+1)^2$ with $\|\hat{A}_\lambda - \hat{A}_\infty\|_{\text{low}} = o(\lambda^{-1/2}/\sqrt{\log \lambda})$. Equivalently: the total positivity of the Loewner Weil matrix is preserved under ground-state renormalization.

The reduction is sharp in the converse direction: for a datum violating RH the local error *cannot* vanish, and this is unconditional.

Proposition 4.16 (The discriminant, converse direction). *Let f be an arithmetic datum whose completed function Ξ_f has a zero $z_0 = x_0 + i\eta_0$ off the real axis ($\eta_0 \neq 0$), and assume the real-axis*

convergence $\widehat{\xi}_{f,\lambda}(x_0) \rightarrow \Xi_f(x_0)$. Since each $\widehat{\xi}_{f,\lambda}$ has only real zeros (Theorem 3.1), a real-rooted entire function of genus ≤ 1 satisfies $|g(x+iy)| \geq |g(x)|$, whence

$$\sup_{|\operatorname{Im} z| \leq |\eta_0|} |\widehat{\xi}_{f,\lambda}(z) - \Xi_f(z)| \geq |\widehat{\xi}_{f,\lambda}(z_0)| \geq |\widehat{\xi}_{f,\lambda}(x_0)| \xrightarrow{\lambda \rightarrow \infty} |\Xi_f(x_0)| > 0.$$

Thus the off-axis local error is bounded below by the explicit constant $|\Xi_f(x_0)| > 0$: a real-rooted approximant cannot converge to a function with a complex zero.

Proof. $\Xi_f(z_0) = 0$ makes the error at z_0 equal to $|\widehat{\xi}_{f,\lambda}(z_0)|$; the real-rooted monotonicity $|g(x+iy)|^2 = \prod_n ((x-r_n)^2 + y^2) \cdots \geq |g(x)|^2$ (real roots r_n) gives $|\widehat{\xi}_{f,\lambda}(z_0)| \geq |\widehat{\xi}_{f,\lambda}(x_0)|$; and x_0 , the real projection of an off-line zero, is not an on-line zero, so $\Xi_f(x_0) \neq 0$. $\square$

Remark 4.17 (Why the gap is the right one: the ζ /DH discriminant). The discriminating property is exactly the sign of the arithmetic part of $d\mu_W$. For ζ it is sign-definite ($\Lambda(n) \geq 0$); for the Davenport–Heilbronn function L_{DH} (which violates RH) the coefficients change sign, Perron–Frobenius fails, and Proposition 4.16 applies at its off-line zero $\frac{1}{2} + 0.3085 + i 85.699$. The two sides match the numerics exactly: with the sup-norm observable the ground spectral function resolves the zeros of ζ to $\eta \sim 10^{-13}$, but never those of L_{DH} below $\eta \sim 10^{-2}$ [6]. Thus ε_{loc} separates the two cases precisely where $\Lambda(n) \geq 0$ does—and the RH-false side is now a theorem (Proposition 4.16), evidence that Problem 4.15 is the genuine residue of RH, not an artifact.

5. LINKS 3 AND 4: HURWITZ AND THE LAGUERRE–PÓLYA CRITERION

Theorem 5.1 (Link 3, Hurwitz). *If $\widehat{\xi}_\lambda \rightarrow \Xi$ uniformly on compact subsets of a strip $|\operatorname{Im} z| \leq \delta$ and each $\widehat{\xi}_\lambda \in \mathcal{LP}$ (Theorem 3.1), then Ξ has no zeros with $0 < |\operatorname{Im} z| \leq \delta$; with $\delta = \frac{1}{2}$ (Theorem 4.6) this gives $\Xi \in \mathcal{LP}$.*

Proof. A zero of Ξ with $\operatorname{Im} z \neq 0$ would be the locally uniform limit of zeros of $\widehat{\xi}_\lambda$, which are real—contradiction by Hurwitz’s theorem. $\square$

Theorem 5.2 (Link 4, classical). $\Xi \in \mathcal{LP} \iff \text{RH}$.

Proof. $\Xi(t) = 0$ iff $\zeta(\frac{1}{2} + it) = 0$ among nontrivial zeros, and Ξ real-rooted means all such t are real, i.e. all nontrivial zeros lie on $\operatorname{Re} s = \frac{1}{2}$ [2]. $\square$

6. ASSEMBLY

Theorem 6.1 (Conditional proof of RH). *Assume Problem 4.15 (equivalently, that $\|\widehat{A}_\lambda - \widehat{A}_\infty\|_{\text{low}} = o(\lambda^{-1/2}/\sqrt{\log \lambda})$, with Conjecture 4.14). Then RH holds.*

Proof. Problem 4.15 with (6) gives $\varepsilon_{\text{loc}}(\lambda) = o(\lambda^{-1/2})$. By Theorem 4.6, $\widehat{\xi}_\lambda \rightarrow \Xi$ uniformly on $|\operatorname{Im} z| \leq \frac{1}{2}$. Each $\widehat{\xi}_\lambda \in \mathcal{LP}$ (Theorem 3.1), so $\Xi \in \mathcal{LP}$ by Theorem 5.1. By Theorem 5.2, RH. $\square$

7. HONEST ASSESSMENT

What is proved: Links 0, 1, 3, 4 unconditionally (Theorems 3.1, 5.1, 5.2); within Link 2, the N -limit (Theorem 4.1), the strip-uniform Fourier truncation (Theorem 4.2), the off-axis transfer (Lemma 4.4), the reduction to ε_{loc} (Theorem 4.6), the relative stability inequality removing the exponentially small gap (Theorem 4.8), the dissolution of the edge layer (Proposition 4.9), and the collapse to a single operator-convergence rate (Corollary 4.10).

What is not proved: Problem 4.15, and the structural identification Conjecture 4.14 that names its limit. We do not claim either. The contribution is the location of the obstruction: along the CCM chain, RH is equivalent (modulo the named structural identification) to a single statement of operator theory—that the Perron–Frobenius positivity carried by $\Lambda(n) \geq 0$ persists through renormalization by the ground energy and fixes a rigid Dirichlet spectrum. This statement contains

no zeros, no Hurwitz, no approximation theory, and no remaining number-theoretic content beyond the sign $\Lambda(n) \geq 0$; it is a question of total positivity for a renormalized Loewner operator. The microscope is now on the right place: the difference between ζ and the Davenport–Heilbronn counterexample has been reduced, through four closed links and a fully developed fifth, to whether one inequality of signs survives one renormalization.

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